1 Conceptual Model

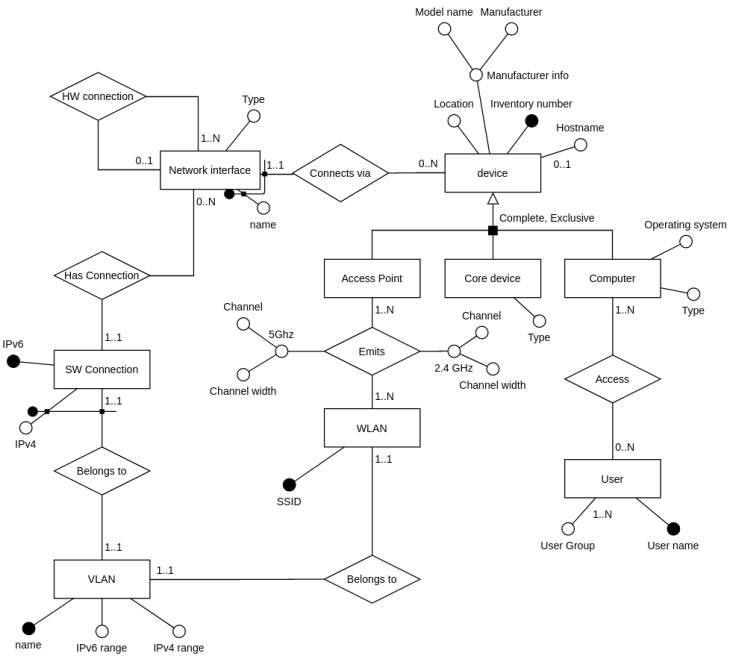

Figure 1: ER diagram

2 Relational Model

Device(<u>invNo</u>, location, hostname, modelName, manufacturer)

Computer(<u>device</u>, os, type)
FK: (device) $\subseteq$ Device(invNo)

User(<u>username</u>)

UserGroup(<u>user, name</u>)
FK: (user) $\subseteq$ User(username)

Access(<u>computer, user</u>)
FK: (computer) $\subseteq$ Computer(device)
FK: (user) $\subseteq$ User(username)

CoreDevice(<u>device</u>, type)
FK: (device) $\subseteq$ Device(invNo)

AccessPoint(<u>device</u>)
FK: (device) $\subseteq$ Device(invNo)

VLAN(<u>name</u>, IPv4Range, IPv6Range)

WLAN(<u>ssid</u>, vlan)
FK: (vlan) $\subseteq$ VLAN(name)

APEmits(<u>ap, network</u>, channel5Ghz, width5Ghz, channel2Ghz, width2Ghz)
FK: (ap) $\subseteq$ AccessPoint(device)
FK: (network) $\subseteq$ WLAN(ssid)

Interface(<u>device, name</u>, type)
FK: (device) $\subseteq$ Device(invNo)

Source can have multiple targets, but target is connected only to one source
HWConnection(<u>sourceName, sourceDevice, targetName, targetDevice</u>)
FK: (sourceName, sourceDevice) $\subseteq$ Interface(name, device)
FK: (targetName, targetDevice) $\subseteq$ Interface(name, device)

SWConnection(<u>ipv6</u>, <u>ipv4, vlan</u>, interface, device)
FK: (interface, device) $\subseteq$ Interface(name, device)
FK: (vlan) $\subseteq$ VLAN(name)

3 SQL

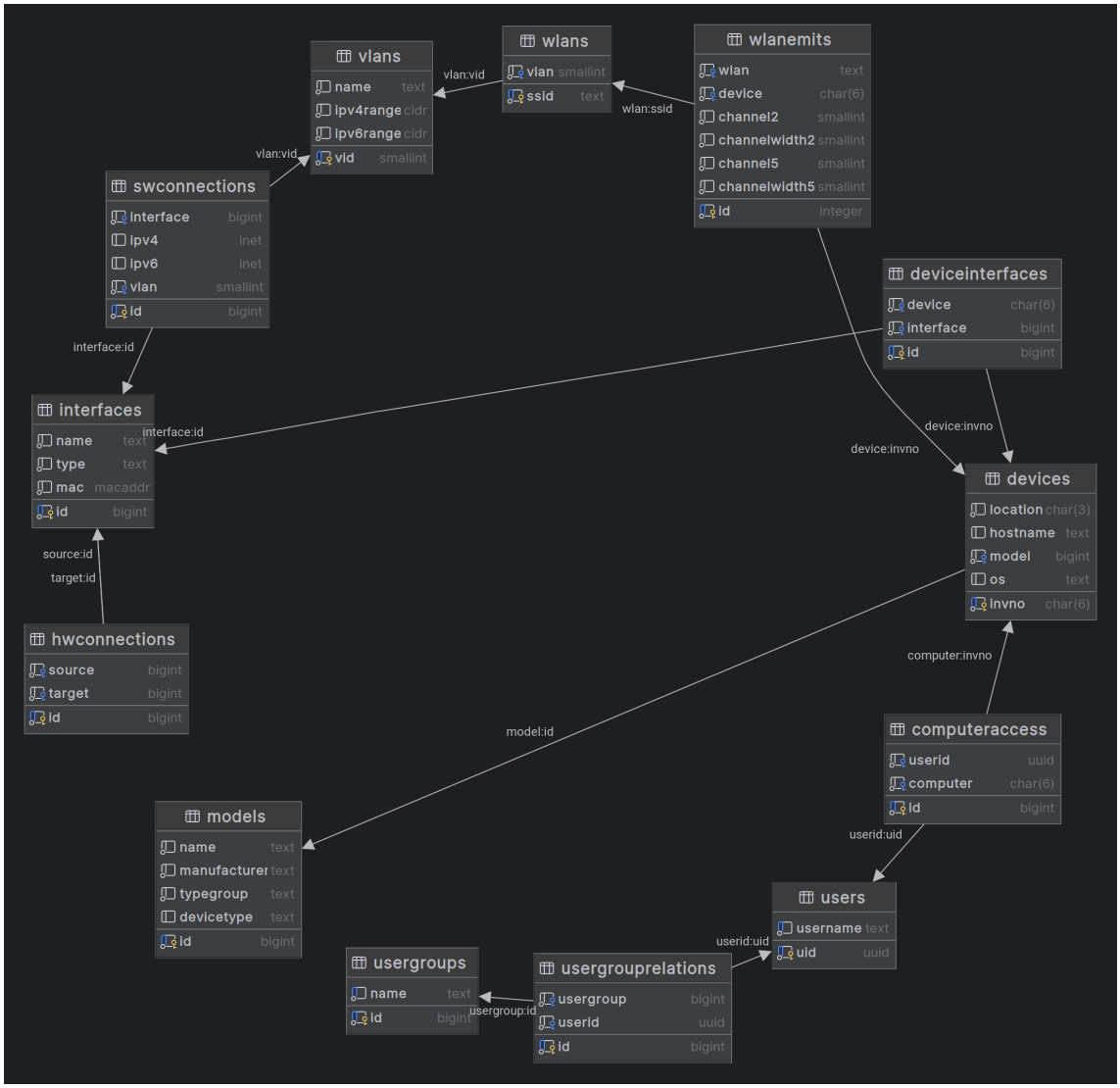

Figure 2: DB diagram

3.1 Creation

```
CREATE TABLE models
(
    id            BIGSERIAL,
    name          TEXT NOT NULL,
    manufacturer  TEXT NOT NULL,
    typeGroup     TEXT NOT NULL DEFAULT 'computer',
    deviceType    TEXT,
    CONSTRAINT Model_pk PRIMARY KEY ("id"),
```

```
    CONSTRAINT Model_typeGroup_values CHECK (
        typeGroup IN ('computer', 'ap', 'core')
        )
);

CREATE TABLE devices
(
    invNo    CHAR(6),
    location CHAR(3)    NOT NULL,
    hostname TEXT,
    model    BIGSERIAL NOT NULL,
    os       TEXT,
    CONSTRAINT Device_pk PRIMARY KEY (invNo),
    CONSTRAINT Device_model FOREIGN KEY (model)
        REFERENCES models (id) ON DELETE SET NULL
);

CREATE TABLE vlans
(
    vid        SMALLINT,
    name       TEXT NOT NULL,
    ipv4range  CIDR NOT NULL,
    ipv6range  CIDR NOT NULL,
    CONSTRAINT Vlan_pk PRIMARY KEY (vid),
    CONSTRAINT Vlan_vid CHECK ( (vid > 0) AND (vid < 4095) )
);

CREATE TABLE wlans
(
    ssid TEXT,
    vlan SMALLINT NOT NULL,
    CONSTRAINT Wlan_pk PRIMARY KEY (ssid),
    CONSTRAINT Wlan_fk_ssid FOREIGN KEY (vlan)
        REFERENCES vlans (vid) ON DELETE SET NULL
);

CREATE TABLE wlanEmits
(
    id              SERIAL,
    wlan            TEXT      NOT NULL,
    device          CHAR(6)   NOT NULL,
    channel2        SMALLINT NOT NULL,
    channelWidth2   SMALLINT NOT NULL,
    channel5        SMALLINT NOT NULL,
    channelWidth5   SMALLINT NOT NULL,
    CONSTRAINT WlanEmits_pk PRIMARY KEY (id),
    CONSTRAINT WlanEmits_wlan FOREIGN KEY (wlan)
        REFERENCES wlans (ssid) ON DELETE CASCADE,
    CONSTRAINT WlanEmits_device FOREIGN KEY (device)
        REFERENCES devices (invNo) ON DELETE CASCADE,
```

```
    CONSTRAINT WlanEmits_channel2 CHECK (
            channel2 IN (
                        1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13
            )
        ),
    CONSTRAINT WlanEmits_chanelWidth2 CHECK (
        channelWidth2 IN (20, 40)
        ),
    CONSTRAINT WlanEmits_channel5 CHECK (
            channel5 IN (
                        32, 34, 36, 38, 40, 42, 44, 46,
                        48, 50, 52, 54, 56, 58, 60, 62,
                        64, 68, 96, 100, 102, 104, 106,
                        108, 110, 112, 114, 116, 118,
                        120, 122, 124, 126, 128, 132,
                        134, 136, 138, 140, 142, 144,
                        149, 151, 153, 155, 157, 159,
                        161, 165, 167, 169, 171,
                        173
            )
        ),
    CONSTRAINT WlanEmits_channelWidth5 CHECK (
        channelWidth5 IN (20, 40, 80)
        )
);

CREATE TABLE interfaces
(
    id   BIGSERIAL,
    name TEXT    NOT NULL,
    type TEXT    NOT NULL DEFAULT 'ethernet',
    mac  MACADDR NOT NULL,
    CONSTRAINT Interface_pk PRIMARY KEY (id),
    CONSTRAINT Interface_type CHECK (
            type IN ('ethernet', 'radio', 'sfp+', 'qsfp+')
        )
);

CREATE TABLE swConnections
(
    id        BIGSERIAL,
    interface BIGSERIAL NOT NULL,
    ipv4      INET,
    ipv6      INET,
    vlan      SMALLINT  NOT NULL,
    CONSTRAINT SwConnection_pk PRIMARY KEY (id),
    CONSTRAINT SwConnection_vlan FOREIGN KEY (vlan)
        REFERENCES vlans (vid) ON DELETE CASCADE,
    CONSTRAINT SwConnection_interface FOREIGN KEY (interface)
        REFERENCES interfaces (id) ON DELETE CASCADE,
```

```
    CONSTRAINT SwConnection_ip CHECK ( (ipv4 IS NOT NULL) OR
                                       (ipv6 IS NOT NULL) )
);

CREATE TABLE hwConnections
(
    id     BIGSERIAL,
    source BIGSERIAL NOT NULL,
    target BIGSERIAL NOT NULL,
    CONSTRAINT HwConnection_pk PRIMARY KEY (id),
    CONSTRAINT HwConnection_source FOREIGN KEY (source)
        REFERENCES interfaces (id) ON DELETE CASCADE,
    CONSTRAINT HwConnection_target FOREIGN KEY (target)
        REFERENCES interfaces (id) ON DELETE CASCADE
);

CREATE TABLE deviceInterfaces
(
    id        BIGSERIAL,
    device    CHAR(6)   NOT NULL,
    interface BIGSERIAL NOT NULL,
    CONSTRAINT DeviceInterfaces_pk PRIMARY KEY (id),
    CONSTRAINT DeviceInterfaces_device FOREIGN KEY (device)
        REFERENCES devices (invNo) ON DELETE CASCADE,
    CONSTRAINT DeviceInterfaces_interface FOREIGN KEY (interface)
        REFERENCES interfaces (id) ON DELETE CASCADE
);

CREATE TABLE users
(
    uid      UUID,
    username TEXT NOT NULL UNIQUE,
    CONSTRAINT User_pk PRIMARY KEY (uid)
);

CREATE TABLE computerAccess
(
    id       BIGSERIAL,
    userId   UUID     NOT NULL,
    computer CHAR(6) NOT NULL,
    CONSTRAINT ComputerAccess_pk PRIMARY KEY (id),
    CONSTRAINT ComputerAccess_userId FOREIGN KEY (userId)
        REFERENCES users (uid) ON DELETE CASCADE,
    CONSTRAINT ComputerAccess_computer FOREIGN KEY ("computer")
        REFERENCES devices (invNo) ON DELETE CASCADE
);

CREATE TABLE userGroups
(
    id    BIGSERIAL,
```

```
    name TEXT NOT NULL UNIQUE,
    CONSTRAINT UserGroup_pk PRIMARY KEY (id)
);

CREATE TABLE userGroupRelations
(
    id          BIGSERIAL,
    userGroup   BIGSERIAL NOT NULL,
    userId      UUID      NOT NULL,
    CONSTRAINT UserGroupRelations_pk PRIMARY KEY (id),
    CONSTRAINT UserGroupRelations_userGroup FOREIGN KEY (userGroup)
        REFERENCES userGroups (id) ON DELETE CASCADE,
    CONSTRAINT UserGroupRelations_userId FOREIGN KEY (userId)
        REFERENCES users (uid) ON DELETE CASCADE
);
```

3.2 Selection

```
SELECT *
FROM swconnections
        RIGHT OUTER JOIN vlans
                        ON (swconnections.ipv4 <<= vlans.ipv4range);

SELECT *
FROM devices
        JOIN models
            ON (devices.model = models.id);

SELECT *
FROM swconnections
WHERE ipv4 = '192.168.32.1';

SELECT COUNT(DISTINCT channel2) AS channels2,
       count(DISTINCT channel5) AS channels5
FROM wlanemits;

SELECT *
FROM models
ORDER BY manufacturer;

SELECT *, MASKLEN(ipv6)
FROM swconnections
WHERE (MASKLEN( ipv6 ) > (
    SELECT AVG(MASKLEN( ipv6 ) ) FROM swconnections )
) ;
```